

LITHALETHU YOTSI, CMSE®

AUTOMATION ENGINEER

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SUMMARY

With over eleven years of hands-on experience spanning heavy manufacturing (tyre production), the automotive sector, and mining engineering services, I possess a robust skill set in automation and industrial systems. My expertise encompasses a large portion of the lifecycle of automation projects, from conceptualization, programming, commissioning to handover. Commissioning servo, DC, and AC drives, to implementing sophisticated shop floor monitoring systems using OPC UA, SDKs, and database integration. I am proficient in programming Siemens, Allan Bradley and Beckhoff PLC/RTU/DCS systems, developing HMI and SCADA visualizations with WinCC and Wonderware InTouch, and programming Fanuc robots. Furthermore, I am adept at troubleshooting pneumatic and mechanical systems, designing and interpreting AutoCAD and E-PLAN schematics, and effectively coordinating projects with contractors. My collaborative approach and ability to work seamlessly within cross-functional teams make me an asset for any organization seeking a seasoned automation professional.

WORK EXPERIENCE

04/2025	**Managing Director & Automation Specialist** Mechatrolix
09/2023 – 03/2025	**Automation Engineer** Detnet
07/2022 – 08/2023	**Automation Engineer** Isuzu Motors South Africa
08/2021 – 06/2022	**Maintenance Engineer** Volkswagen SA
01/2017 – 07/2021	**Controls And Drives Technician** Continental Tyres SA
06/2015 – 01/2017	**Controls And Drives Trainee** Continental Tyres SA
03/2015 – 07/2015	**Quality Trainee** Continental Tyres SA
07/2014	**Trainee lathe operator (Vacation training)** Ashes Clutch & Brake

EDUCATION

2011 – 2018	**BEng Mechatronics** Nelson Mandela University Gap Year (2017)
2006 – 2010	**Selborne College, East London**
2004 – 2005	**Komga Junior, Komga**
2000 – 2003	**River of Life Christian School, Cathcart**
1997 – 1999	**Ndyebo Ntsaluba JSS, Gqogqorha**

COMPETENCIES

- Electrical, Control and Instrumentation Solution Specification
- IEC Electrical Schematics and Industrial Control Panel Design
- PLC/RTU Configuration and Programming
- Comprehensive OT/Supervisory, Automation and Field Layer Detailed Design
- P&ID Interpretation
- SCADA Development
- VSD Sizing, Selection and Commissioning
- Machine Electrical and Mechanical Design
- Higher level plant support for low voltage electrical applications and software

APPLICATIONS

- Autodesk Inventor
- Siemens s7 & WinCC
- Wonderware InTouch & Archestra
- Siemens PCS7
- Beckhoff TwinCAT 2 & 3
- EPLAN P8 and Pro Panel
- MS Office
- SQL
- Visual Studio
- AB Studio 5000 & FactoryTalk
- Solidworks
- Siemens s5
- Siemens STARTER/Startdrive
- TIA Portal & WinCC (Comfort/Advanced/Professional/Unified)
- Codesys
- Omron CX-Programmer
- MS Access
- MATLAB
- Ignition SCADA
- FANUC Roboguide

AWARDS

Outstanding Achievement - ISUZU Motors

Awarded for developing a proprietary shop-floor monitoring application for the Body Shop, designed to monitor production bottlenecks, engineering breakdowns, and Overall Equipment Effectiveness (OEE). The software operated with minimal human intervention, requiring only fault acknowledgements and operator logins. Multiple thin-client terminals were strategically installed near operator stations and robotic cells to enable human–machine interfacing (HMI) and to accurately log faults, cycle overruns, and troubleshooting guidance. The system generated automated performance reports and provided a live business intelligence (BI) dashboard accessible remotely.
The application was developed using Kepware OPC UA DataBridge, an OPC UA-to-Windows SDK, WPF (Windows Presentation Foundation) forms and Blazor with .NET backend, and Microsoft SQL Server for backend data management and hosting.

COURSES

02/2019 – 02/2019
Port Elizabeth

Siemens SCT-S7NETSIA
SITRAIN, Nelson Mandela University
- Introduction to PROFINET
- Introduction to PROFIBUS

01/2019 – 01/2019
Port Elizabeth

Siemens ST-7PRO1
SITRAIN, Nelson Mandela University
- S7 Product Overview
- S7-300/400 PLC Hardware Configuration
- S7-300/400 PLC Program structuring
- Binary, Digital Operations
- DBs and OBs
- FCs and FBs
- Integrating and Commissioning with a Drive and STARTER
- S7-300/400 Logic creation in LAD, GRAPH, SCL and FBD
- Introduction HMI Development

02/2020 – 02/2020
Port Elizabeth

Siemens TIA-PRO1
SITRAIN, Nelson Mandela University
- S7 Product Overview
- S7-1200/1500 Device and Network Configuration
- S7-1200/1500 Program Structuring
- Binary, Digital Operations
- DBs and OBs
- FCs and FBs
- Intergrating and Commissioning with a Drive and StartDrive
- S7-1200/1500 Logic creation in LAD, GRAPH, SCL and FBD

11/2023 – 11/2023
Randburg

Introduction to Beckhoff Programming
Beckhoff SA
- Introduction to TwinCAT Programming, including safety, motion control and HMI design

04/2025 – 04/2025
Johannesburg

Robot Programming and Operation
Fanuc Academy
- Configuration of Fanuc Robot
- Program and Motion Instructions
- Practice for Teaching/Operation
- Peripheral Device Control Instructions
- Control Instructions for End Effector
- Control Instructions for Hand
- Macro Commands
- Co-ordinate Systems

01/2026 – 01/2026
Online, USA

CMSE® – Certified Machinery Safety Expert
TÜV NORD
- Machinery Risk Assessment and Risk Reduction
- Hazard Identification and Risk Evaluation
- Machinery Safety Legislation and Standards
- CE Conformity Assessment
- Safety Requirements Specification (SRS)
- Functional Safety of Control Systems
- Performance Level (PL) Calculation – ISO 13849
- Safety Integrity Level (SIL) Concepts – IEC 62061
- Validation and Verification of Safety Systems
- Machine Guarding and Protective Measures
- Electrical Safety in Machinery (IEC 60204-1 / NFPA 79)
- Lockout / Energy Isolation Safety Principles
- Safety Component Selection and Application
- Emergency Stop and Safety Function Design
- Pneumatic and Hydraulic Safety Systems

★ CONTROLS AND DRIVES TECHNICIAN-CONTINETAL TYRES SA

As a Controls and Drives Technician, I was responsible for the comprehensive management, optimisation, and integrity of all plant control systems and automation infrastructure, encompassing Industrial PCs (IPCs), PLCs, drive systems, and communication networks. My role ensured seamless integration between operational technology (OT) and manufacturing systems while maintaining strict adherence to Continental's global Controls and Drives (C&D) standards.

Core Responsibilities

- **Control Software & Configuration Management**

I oversaw all plant control software—including IPCs, PLCs, and drive communication systems. I executed software modification, ensuring that all software changes were properly authorised, documented, and version-controlled through VersionDog. I issued and retrieved control system software versions while maintaining a fully auditable backup repository.

- **System Commissioning & Integrity**

I led the installation, configuration, and commissioning of IPCs and PLCs in conjunction with contractors, performing comprehensive integrity checks and sign-offs. I coordinated SFI machine integration rollouts, network configuration (IP allocation, host naming), system policy implementation, and third-party access control. I maintained the Engineering Servers, Backup & Restore environments, ensuring application-level integrity for platforms such as FFDACS, Grading, SFI, and MMS.

- **Standards Definition & Compliance**

I defined and enforced plant-specific standards aligned with global Continetal C&D guidelines, covering IPCs, PLCs, drives, HMIs, and SCADA interfaces. I collaborated with Project and R&M Engineers to prepare detailed machine control specifications and conducted acceptance testing prior to commissioning.

- **Control System Upgrades & Long-Term Planning**

I planned and coordinated control system hardware and software upgrades in collaboration with Section Engineers. I developed 3-year upgrade roadmaps covering IPC replacements, control software modernisation, budgeting, and adherence to Continental's engineering standards. I oversaw eCR (engineering change requests) related to system enhancements and hardware replacements.

- **Hardware & Software Expertise**
  - Hardware: IPCs, PLCs, industrial drive systems, communication networks, and machine safety systems.
  - Software: PLC Control configuration, high-level programming, and application support for manufacturing automation.
- **Training & Knowledge Transfer**

I delivered IPC systems, PLC, HMI, VSD training and architectural knowledge to R&M Technicians, in conjunction with the Training Department. I led plant-wide coordination of control activities and knowledge management initiatives.

- **Vision Systems Integration**

  1. I was proficient in the design, commissioning, and integration of industrial vision systems from concept to production deployment.
  2. I designed and fabricated mechanical camera assemblies for optimal modularity and maintainability.
  3. I configured network communication between vision cameras and PLC systems using CP cards, PN/DP couplers, or serial (Visual C#) connections.
  4. I imported and configured GSD files for proper data mapping to the control network.
  5. I implemented uniform lighting systems to maximise image contrast and accuracy.
  6. I performed focus, distance, and field-of-view calibration to achieve ideal pixel density and precision.
  7. I developed vision inspection algorithms (edge, circle, line detection, uniformity, etc.) based on the system's capabilities.
  8. I designed comprehensive HMI interfaces for measurement visualisation, system alarms, and operator interaction.
  9. I validated system repeatability through structured test and verification procedures before handover.

- **Motion Control & Drives Engineering**

  1. I have extensive experience in commissioning and integrating drive systems across multiple vendor platforms.
  2. I performed mechanical and dynamic calculations for torque, inertia ratios, ramp profiles, and kW sizing.
  3. I integrated safety systems (Siemens F-Safe catalog) to provide motion interlocks and emergency stops.
  4. I configured motor control integration via GSDs, telegrams, or analog scaling to automation controllers.
  5. I interfaced load cells and analog sensors for precision measurement in delicate handling applications.
  6. I commissioned and controlled pneumatic motion systems and integrated them into central PLC logic

★ MAINTENANCE ENGINEER – VOLKSWAGEN SOUTH AFRICA (VWSA)

As a Maintenance Engineer at Volkswagen South Africa, I was responsible for ensuring the reliability, maintainability, and continuous improvement of plant equipment and infrastructure. My role encompassed multi-disciplinary technical support across electrical, mechanical, structural, and automation systems, while aligning all activities with the organisation's production, safety, and sustainability objectives.

Core Responsibilities

- **Technical Support & Problem Solving**

I provided multi-disciplinary engineering support to maintenance supervisors and departments, addressing complex issues across electrical, mechanical, structural, and process systems. I utilised advanced diagnostic methods to analyse root causes and implemented corrective and preventive actions to ensure sustainable plant performance.

- **Stakeholder Collaboration & Efficiency Improvement**

I liaised with key stakeholders—including Production, Maintenance, Process, Quality, MPE, and Logistics teams—to improve operational performance indicators such as Overall Plant Reliability (OPR) and Downtime Reduction Rate (DRR). I worked closely with cross-functional teams to ensure technical solutions aligned with production targets and long-term plant objectives.

- **Technology Advancement & Sustainable Reliability**

I remained up to date with the latest industry technologies and process advancements to drive innovative and sustainable maintenance strategies. My focus was on improving facility robustness, reliability, and maintainability, while integrating new technologies into existing systems.

- **Safety & Design Compliance**

I ensured that all new and existing installations adhered to safe design principles and engineering best practices. I participated in design reviews, risk assessments, and commissioning activities to guarantee compliance with safety and maintenance standards.

- **Breakdown Management & Preventive Maintenance**

I supported and, where required, led breakdown escalations, coordinating rapid technical response to restore operations. I developed and implemented preventive maintenance strategies, documenting root causes and driving follow-through actions to prevent recurrence.

- **Spare Parts & Inventory Management**

I facilitated the procurement of critical spares, managed stock-holding levels, and optimised stores operations to support maintenance reliability and reduce downtime.

- **Process & System Improvement**

I developed and implemented maintenance systems and procedures aimed at improving efficiency, traceability, and compliance. I promoted continuous improvement through structured documentation, metrics tracking, and lessons learned.

- **Sustainability & Energy Efficiency (Think Blue Factory)**

I actively contributed to the Think Blue Factory (TBF) initiative by compiling annual reports, developing sustainability targets, and managing input for energy-saving projects. I participated in bi-weekly energy team meetings, monitoring progress on initiatives that improved overall energy efficiency.

- **Technical Proficiencies**

Automation & Drives

- Proficient in SEW and Siemens Drive Systems, including parameterisation, commissioning, and diagnostics.
- Skilled in industrial communication networks such as Profibus, Profinet, and OPC for data exchange and system integration.

PLC & Robotics Programming

- Advanced proficiency in Siemens PLC and F-PLC programming, including fault diagnostics, logic optimisation, and system commissioning.
- Experienced in robotic programming for Fanuc R-Series and KUKA KRC2/KRC controllers for automation and material handling applications.

HMI & SCADA Development

- Developed and customised HMI interfaces for production lines, providing intuitive process visualisation, alarm management, and operator control functionalities.

Professional Summary

In this capacity, I combined technical expertise, cross-departmental coordination, and process optimisation to strengthen plant reliability and performance. My role demanded a strong balance between hands-on engineering capability and strategic maintenance planning, ensuring compliance, sustainability, and continuous improvement within one of South Africa's most advanced manufacturing environments.

★ AUTOMATION ENGINEER – IMSAF (ISUZU MOTORS SA)

As an Automation Engineer at Isuzu Motors South Africa, I was responsible for the design, analysis, troubleshooting, and optimisation of control systems across the plant, encompassing PLC, HMI, SCADA, and robotic automation platforms. My role integrated deep technical problem-solving with engineering leadership to enhance production efficiency, safety, and reliability across manufacturing operations.

Core Responsibilities

- **Control Systems Analysis & Program Development**

I developed and analysed PLC, HMI, SCADA, and robot programs for the purpose of fault finding, modification, and design enhancement. I implemented error-proofing logic and developed program improvements to optimise machinery performance, cycle time, and reliability.

- **Troubleshooting & Technical Support**

I provided hands-on support in resolving program logic, electrical, and mechanical faults on production machinery to minimise downtime. I collaborated closely with maintenance teams to support production continuity, equipment safety, and product quality.

- **Technical Documentation & Interpretation**

I designed and interpreted electrical schematics, technical manuals, PLC ladder/SCL logic, and robot programs to systematically troubleshoot and restore machine operation.

- **Breakdown Assistance & Root Cause Analysis**

I assisted artisans and technicians during breakdowns, providing advanced fault diagnostics. I conducted root cause analyses (RCA) and developed preventive countermeasures to eliminate recurring issues.

- **Training & Skill Development**

I developed and facilitated technical training programs for Graduates-in-Training (GITs), In-Service Trainees (ISTs), artisans, and technicians—enhancing their understanding of automation systems, diagnostics, and control software.

- **Spare Parts & Resource Management**

I established and maintained a strategic spare parts inventory, ensuring the continuous availability of critical automation components. This proactive approach reduced downtime and improved equipment readiness.

- **Project Support & Validation**

I supported contractors and integrators during new system installations and performed validation, commissioning, and acceptance testing to ensure compliance with Isuzu and statutory requirements.

- **Efficiency & Process Improvement**

I identified and implemented process optimisation initiatives, focusing on efficient use of utilities, materials, and energy resources to improve overall equipment effectiveness (OEE) and cost performance.

- **Budget & Cost Control**

I maintained strict cost management of CAPEX and OPEX budgets, ensuring projects were delivered within scope, schedule, and budget constraints.

Project & Engineering Responsibilities

- **Automation & Tooling Development**

I was responsible for the planning, design, manufacturing, installation, and commissioning of all controls and automation elements related to tooling, production equipment, and facilities—supporting new model introductions, IT interface refreshes, and process upgrades.

- **Safety & Compliance**

I ensured all equipment designs promoted safe work practices and met Isuzu global standards as well as South African safety, health, and environmental regulations. I contributed to hazard assessments, safety validations, and commissioning approvals.

- **Team Leadership & Contractor Management**

I managed and coordinated both internal teams and external contractors to achieve program milestones, ensuring a focus on safety, quality, delivery lead times, and performance targets.

- **Feasibility & Cost Studies**

I conducted technical feasibility assessments and cost estimation studies for proposed automation and controls projects—evaluating scope, ROI, and resource requirements for management approval.

- **Design & Stakeholder Reviews**

I facilitated equipment and tooling design reviews and buyoff meetings with all relevant stakeholders—including production, maintenance, quality, and process engineering—to ensure alignment and functionality prior to commissioning.

- **Engineering Change Implementation**

I managed and implemented engineering changes to control systems, ensuring accurate documentation, revision control, and smooth transition into production without affecting process integrity.

Professional Summary

In this role, I combined automation engineering, system integration, and project management expertise to deliver reliable, safe, and optimised control solutions for Isuzu's manufacturing operations. My responsibilities spanned from advanced fault diagnostics and root cause prevention to full lifecycle management of automation projects, ensuring compliance with both Isuzu global standards and South African engineering regulations.

★ AUTOMATION ENGINEER – DETNET SOUTH AFRICA

As an Automation Engineer at DetNet South Africa, I was responsible for leading the design, optimisation, and integration of automation systems within the company's manufacturing environment. My role focused on continuous improvement, technology innovation, and cross-functional collaboration to enhance production efficiency, product quality, and long-term manufacturing sustainability.

Core Responsibilities

- **Process Improvement & Equipment Optimisation**

I identified areas of improvement in existing manufacturing equipment and processes, driving the implementation of replacement technologies and upgraded systems to improve reliability, efficiency, and performance.

- **Technology & Standards Development**

I continuously evaluated and introduced new technologies, processes, and standards that could enhance DetNet's supply chain and manufacturing capabilities. I played an active role in standardising automation engineering practices, ensuring a consistent framework for system design, commissioning, and maintenance.

- **Project Leadership & System Development**

I took a technical lead role in the development of new automation systems within cross-functional project teams. I contributed to the design, programming, and integration of control systems, ensuring seamless implementation within existing production environments.

- **Outsourced Project Support & Oversight**

I provided technical and managerial support for outsourced automation projects, assisting in the creation of User Requirement Specifications (URS), offering technical guidance, and performing commissioning oversight. I ensured that all outsourced systems were fully integrated, tested, and compliant with DetNet's manufacturing standards.

- **Product Development Support**

I provided manufacturing and automation insight during new product development and major product change initiatives. My input influenced product and process design early in the product life cycle, ensuring manufacturability, scalability, and automation readiness.

- **Collaboration with R&D and Industrialisation Teams**

I worked closely with Research and Development (R&D) teams to stay informed on new product innovations and provided support during the industrialisation process—translating laboratory concepts into full-scale manufacturing solutions.

- **Automation & Manufacturing Standards**

I developed and supported automation and manufacturing standards for engineering projects, including:

1. Software development and version control guidelines
2. Documentation and project planning templates
3. Standardisation of parts, control components, and system architectures
4. Commissioning and acceptance procedures

These standards ensured consistency, quality, and traceability across all automation projects.

- **Legacy System Support**

I provided technical assistance and maintenance support for legacy automation systems, ensuring compatibility, reliability, and smooth integration with newer technologies.

- **Strategic Manufacturing Contribution**

I contributed to long-term manufacturing strategies, offering technical insights to guide investment, scalability, and digital transformation initiatives within DetNet's production facilities.

- **Knowledge Transfer & Continuous Development**

1. I actively shared technical knowledge and best practices with other team members to enhance the organisation's manufacturing technical capability.
2. I maintained a strong focus on continuous technical development, expanding my expertise in automation systems, industrial communication, and control architecture.
3. I engaged with senior manufacturing personnel to align automation initiatives with strategic goals and to strengthen interdepartmental communication and collaboration.
4. I implemented project communication frameworks and cross-department planning strategies, fostering better integration between engineering, production, and R&D teams.

Professional Summary

In this role, I combined technical leadership, process engineering, and automation design expertise to advance DetNet's manufacturing excellence. My work ensured that automation initiatives not only supported day-to-day production efficiency but also aligned with the company's long-term vision for innovation, standardisation, and sustainable manufacturing growth.

★ AUTOMATION SPECIALIST - MECHATROLIX

05/2025 – 10/2025
Ruvuma, Tanzania

Mkuju River Uranium Pilot Plant

PLC/SCADA/ASI bus Commissioning and VSD Parameterization

As an Automation and Commissioning Engineer, I was responsible for the complete PLC, SCADA, AS-i Bus, and VSD parameterization for the Mkuju River Uranium Pilot Plant in Tanzania. The project was executed under Mantra Tanzania (ROSATOM) as the main client, with PCS Global serving as the primary automation subcontractor. The pilot facility was designed for a 6 tons/hour processing throughput, serving as a pre-commercial validation unit for the full-scale uranium processing plant.

My role encompassed end-to-end engineering, integration, programming, commissioning, and documentation of the plant's automation systems, covering more than 15 operational areas, from ore handling to water treatment and reagent dosing.

Core Responsibilities

1. PLC Programming and Control Architecture

I was responsible for the PLC programming, hardware configuration, and commissioning of the plant's central control system based on the Siemens S7-1500 PLC platform.

- Developed comprehensive control logic using a hybrid of STL (Statement List), SCL (Structured Control Language), LAD (Ladder Logic), and GRAPH (Sequential Function Chart) programming languages to achieve modular and transparent process control.
- Implemented modular function blocks and diagnostic routines to standardize interlocking, drive control, analog scaling, and alarm management across plant areas.
- Designed control strategies for 34 variable speed drives (VSDs) and 42 direct-on-line (DOL) motor starters, ensuring each device was fully interlocked with field instrumentation, permissive signals, and safety circuits.
- Created sequential control logic for multi-stage process units including the ore scrubbing, pre-thickening, leaching, elution, resin-in-pulp, and precipitation circuits, ensuring correct interdependencies and safe state transitions.

2. VSD Parameterization and Integration

The plant's motor control system utilized Nidec Control Techniques C200 and C300 variable speed drives.

- Carried out parameterization of all VSDs, including motor data entry, acceleration/deceleration ramps, I/O offsets and scaling.
- Integrated each drive into the AS-i network.
- Configured local and remote control modes, ensuring safe handover between operator panels and SCADA control.
- Performed functional testing under load for conveyors, pumps, agitators, to validate torque, frequency and feedback performance.

3. AS-Interface (AS-i) Bus Network Commissioning

The field instrumentation was largely decentralized using an IFM AS-i Bus network for discrete and analog devices.

- Commissioned the AS-i topology, segmenting the network per process area to ensure isolation and fault containment.
- Integrated valves, limit switches, proximity sensors and instruments via IFM AC1402 gateways and AS-i modules to the Siemens PLC.
- Configured addressing, diagnostics, and slave mapping within TIA Portal, ensuring robust data integrity and fast cyclic communication.
- Verified redundancy and fault diagnostics, ensuring that device-level interruptions did not propagate across process segments.

4. SCADA Development and System Integration

Developed and deployed a WinCC Professional SCADA system for plant-wide monitoring and control.

- Configured Engineering Station (ES) and Operator Station (OS) architecture with user access management, automatic backup policies, and redundant data logging.
- Developed over 20 process screens encompassing analog/digital process displays, alarm windows, motor and valve faceplates, and operator dialogs.
- Implemented dynamic graphics, interlock visualization, trend pages, and historical data archiving for process analysis and fault tracking.
- Configured and optimized tagging, area hierarchy, and alarm management according to ISA-18.2 guidelines.
- Integrated Profinet data channels to external Microsoft SQL Server databases using Siemens Industrial Data Bridge, enabling external process data logging, historian replication, and downstream reporting for engineering and management teams.

5. Instrumentation, Valves, and Field Devices

Commissioned all analog and digital field instruments across multiple process areas.

- Configured and scaled analog devices including pressure transmitters, flow meters, pH sensors, level transmitters, and temperature sensors.
- Calibrated control and isolation valves, establishing correct PID loop direction and feedback.
- Developed and tested control logic for digital valves (open/close feedback with interlocks) and analog valves (modulating valves for flow and level control loops).
- Validated instrument diagnostics, ensuring that all transmitters and field devices correctly reported status and faults to the control system.

6. Process Areas Covered

The automation scope covered the entire pilot plant process chain:

- Ore Scrubbing Section
- Pre-Thickening Area
- Acid Atmospheric Leach Area
- Resin-in-Pulp (RIP) Tails and Handling Sections
- Elution Circuit
- Uranium Precipitation Area
- Raw and Process Water Systems

- Water Treatment and Potable Water Plants
- Tails/Scats Transfer Systems
- Excess Water Treatment Area
- Sulphuric Acid Dosing Area
- Hydrogen Peroxide and Sodium Hydroxide Dosing Areas
- Lime Handling System
- Compressed Air Plant

Each area was engineered for full automated sequence control, ensuring safety interlocks, startup sequencing, and process fail-safes were properly validated.

Project Management & Documentation

- Managed punch-list tracking, recording and resolving commissioning issues identified during cold and hot testing phases.
- Maintained progress tracking logs, ensuring all activities met project deadlines and compliance standards.
- Produced as-built documentation including updated PLC I/O lists, network diagrams, electrical schematics and system architecture drawings.
- Delivered complete commissioning and handover documentation, including control narratives, loop check records, and system operation manuals.

Professional Summary

In this project, I executed complete automation lifecycle management—from PLC and SCADA engineering through to commissioning and documentation—for Tanzania's first uranium plant. The role demanded deep technical knowledge of Siemens TIA Portal, WinCC Professional, Profinet/AS-i communications, and VSD integration within a high-integrity process control environment. The project successfully demonstrated automated process control for a 6 t/h uranium extraction pilot facility, establishing a robust and scalable framework for the future commercial uranium plant.

12/2026 – 01/2026
Remote- Saudi Arabia

Optimizing refurbished ship fin stablilizer

Fault finding and optimization for program of dual ship fin stabilizers on a Siemens S5 PLC for a client based in Saudi Arabia.

01/2026 – 03/2026
Kriel, Mpumalanga

Eskom North/South Lime Plant Refurbishment

PLC and HMI Engineering for the refurbished North and South Lime Plants at Kriel power station. This was a brown field project and I was required to develop a program and HMI screens for the new control philosophy which I was given freedom to optimize. The electrical and mechanical infrastructure required maintenance and a new FAT and SAT which I developed the project plan for.

02/2026 – 04/2026
Remote - United States

WinCC Professional <> Odoo Interfacing

Led the full-cycle design and implementation of a bidirectional integration between Siemens WinCC RT Professional and Odoo ERP for automated batch production and recipe management. The project spanned 11 weeks across four structured phases: system discovery and architecture design, inbound integration, outbound integration, and deployment.

Conducted a comprehensive assessment of the existing TIA Portal PLC programs, WinCC configuration, and SQL Server infrastructure before designing the integration architecture. Developed inbound integration enabling automatic transmission of production orders and recipes from Odoo to the Siemens environment, including operator validation workflows and role-based access control on WinCC HMI screens.

Implemented outbound integration capturing real-time weighing data, material consumption, and batch genealogy, with automated inventory reconciliation posted back to Odoo. Developed variance analysis, traceability reporting, and error handling with retry and buffering logic to ensure data integrity during network interruptions.

Delivered full documentation covering system architecture, data flow mapping, operational procedures, and maintenance guides, along with operator and administrator training. The solution was deployed to production with a shadow operation period to validate accuracy before full cutover.

04/2026
Middelburg,Mpumalanga

Waster Water Treatment Plant EC&I specification

Contracted to specify an electrical, control and instrumentation solution for the Blinkpan Wasterwater Treament plant that's being overhauled. I have designed the MCC and PLC control panel and interfaced a power quality measuring system and field instrumentation network using EPLAN and Pro Panel, the control system will run on a Siemens S7-1500 PLC with ET200 distributed I/O stations. Cybersecurity and network hardening are within the scope and SCADA will be developed on WinCC Unified.

REFERENCES

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